Investment Memo: ASTS

AST SpaceMobile, Inc. | Nasdaq: ASTS Memo Date: May 28, 2026 | Analyst: Theory Y Equity Research Sector: Space Infrastructure / Satellite Telecommunications

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1. One-Line Thesis & Valuation Snapshot

AST SpaceMobile is the only company with an FCC-licensed, commercially operational direct-to-device (D2D) satellite broadband network that works with unmodified smartphones — a structural moat in the race to become the wholesale coverage layer beneath every mobile network operator on earth, priced today on a revenue-ramp faith trade that demands disciplined sizing.

Valuation Snapshot (as of August 2025 data; public filings through Q2 2025):

As of mid-2025, ASTS traded in a wide band reflecting competing narratives: the bear case pricing in serial dilution and execution risk against a SpaceX/T-Mobile juggernaut, the bull case pricing in a multi-billion-dollar recurring revenue stream from MNO wholesale contracts. With ~250–280M fully diluted shares outstanding (post-Block 2 equity raises) and a stock oscillating between ~$12–$32 during H1 2025, the implied enterprise value ranged from approximately $3.0B to $8.5B. Against trailing twelve-month revenue of sub-$20M (predominantly grant/government revenue and early commercial trials), any EV/Revenue multiple is a distraction — the market is pricing probability-weighted DCF scenarios on a constellation that does not yet exist at commercial scale.

The more useful framing: Block 1 (5 BlueBird satellites, commercially live as of late 2024) demonstrated technical feasibility with AT&T and Verizon. Block 2 (~60 next-generation satellites planned for 2025–2026 launch cadence) is the revenue inflection. At full Block 2 deployment, AST management has guided toward capacity to serve approximately 20–25M subscribers in the US alone, with carrier MOU economics suggesting wholesale pricing of $5–10/subscriber/month for seamless coverage access. If 10M subscribers are activated at $7/month average wholesale rate across contracted MNOs, that's ~$840M annualized revenue — at 1.5–2.0x that figure's EV implied by mid-range share prices, the market is effectively pricing in 50–60% probability of commercial-scale execution with a mid-cycle revenue multiple of 4–5x. That is not irrationally expensive for a winner-take-most infrastructure position, but it leaves zero margin for delays, dilution, or competitive displacement.

The stock is not a value trade. It is a real-option instrument on satellite infrastructure dominance.

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2. Business Overview & Operations

AST SpaceMobile builds and operates a space-based cellular broadband network designed to provide 4G LTE and 5G NR connectivity directly to standard unmodified mobile handsets — no satellite phone, no specialized modem, no app required. This is the core technical claim that separates AST from every prior satellite telephony venture (Iridium, Globalstar, Ligado) and from SpaceX Starlink's current consumer broadband product.

Technology Architecture

The BlueBird satellite bus uses an exceptionally large phased-array antenna — the largest commercial communications arrays ever deployed in low earth orbit — to concentrate enough downlink energy to reach a standard 3GPP-compliant handset with a sub-0.5W transmitter. Block 1 BlueBirds (launched September 2024, 5 satellites) validated this architecture at 4G speeds in sub-2 GHz licensed spectrum. Block 2 satellites are

designed with larger arrays and higher power budgets targeting 5G NR with multi-channel operation, which will substantially increase per-satellite capacity.

Revenue Model

AST's go-to-market is B2B wholesale, not direct-to-consumer. Mobile network operators license access to AST's satellite coverage layer to offer their subscribers seamless connectivity in geographic areas where terrestrial infrastructure is economically unviable to build. The MNO bears subscriber acquisition, billing, and retention costs; AST receives a wholesale fee per active subscriber or a revenue share. This model has two important operating implications:

1. **Customer concentration is high but credit quality is exceptional.** AT&T and Verizon anchor the US commercial agreements. Internationally, Vodafone Group, Telefónica, Rakuten Mobile, and others have signed commercial agreements covering populations across Europe, Japan, and emerging markets. These are investment-grade counterparties with long-duration contracts and high switching costs once integrated.
2. **AST's cost structure is predominantly satellite CapEx and launch costs**, not a variable cost per subscriber served. Once a satellite is in orbit and its licensed spectrum is cleared, the marginal cost of serving an additional subscriber is negligible — this is the operating leverage story. Fixed costs are the constellation build-out; incremental revenue drops largely to gross margin.

Cost Structure and Operating Leverage

Satellite manufacturing and launch represent ~$150–250M per Block 2 tranche (exact figures not publicly disclosed at the per-satellite level). Operating expenses include ground station infrastructure, spectrum licensing maintenance, R&D for Block 3/next-gen development, and G&A for a ~700-person company (as of 2025 headcount disclosures). EBITDA is deeply negative in the build phase — AST was burning ~$60–90M per quarter in 2024–2025 before meaningful commercial revenue. The operating leverage thesis depends entirely on Block 2 satellites becoming revenue-generating before the balance sheet requires another large equity raise at dilutive prices.

Go-to-Market Status

Block 1 commercial service launched with AT&T and Verizon in late 2024, covering portions of the continental US for beta subscribers. Revenue recognition under these agreements was limited by satellite count and coverage density. Block 2 launch cadence (targeting SpaceX Falcon 9 rideshares through 2025) is the single most important near-term execution variable. Until Block 2 is substantially deployed, AST has a proof-of-concept product, not a commercial-scale service.

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3. Alternative Data & Macro Signals

The Connectivity Gap TAM is Real and Growing

ITU and GSMA data consistently show approximately 3–4 billion people in areas where mobile broadband is technically unavailable or economically underserved by terrestrial networks. The economics of towers in low-population-density or complex terrain geographies do not pencil for MNOs — return on invested capital for rural tower builds is typically below the 8–10% WACC threshold at population densities below ~500 people/sq km. Satellite coverage-as-a-service to MNOs removes that CapEx burden entirely. This is not a speculative market — it is a documented failure of terrestrial economics that satellite infrastructure can address at meaningfully lower per-unit cost.

MNO CapEx Discipline Favors the AST Model

Post-2022, major global MNOs have been under sustained pressure to improve free cash flow and reduce leverage after heavy 5G spectrum auction and deployment spending. AT&T's $20B+ 5G midband buildout and debt load, combined with activist pressure on capital allocation, creates a structural preference for coverage-gap solutions that don't require additional CapEx on their balance sheet. AST's wholesale model is essentially off-balance-sheet rural coverage for its carrier partners — economically attractive precisely because MNO capex cycles are constrained.

Government and Defense Demand as an Underappreciated Signal

The US Space Development Agency, DARPA, and Department of Defense have publicly funded AST's technology development through multiple contracts totaling over $150M in aggregate through 2025. This is not charity — DoD has a direct operational interest in resilient, non-terrestrial communications for troops and assets in denied or degraded environments. Government revenue provides a floor under the income statement during commercial ramp and validates the technology at the highest due-diligence standard. No commercial customer scrutinizes a satellite communication architecture more thoroughly than the Pentagon.

Spectrum Licensing as a Competitive Moat

AST holds FCC Part 25 satellite licenses to operate in licensed carrier spectrum (rather than ISM/unlicensed bands). This is structurally significant: because AST transmits in spectrum already licensed to its MNO partners (AT&T and Verizon spectrum, Vodafone spectrum, etc.), there is no new spectrum risk and no interference concern for the end-user device. SpaceX's T-Mobile D2D partnership operates in T-Mobile's licensed spectrum under a different regulatory structure, but SpaceX's multi-operator ambitions face a more complex spectrum-sharing puzzle. AST's single-operator-per-market model aligns incentives cleanly.

The SpaceX Competitive Threat: Competitive but Not Displacement

The T-Mobile/SpaceX D2D service (Beta launched 2024, commercial rollout 2025) is the most-cited bear catalyst. Evaluating it accurately requires separating the US domestic market (where T-Mobile/SpaceX competes directly with AST/AT&T and AST/Verizon) from the international market (where SpaceX has no comparable carrier agreements outside a handful of markets). In the US, T-Mobile/SpaceX has messaging and initial data service but has not demonstrated the multi-megabit 5G NR throughput that AST's Block 2 targets. For enterprise and government use cases requiring real broadband capability — not just SMS/low-bitrate voice — AST's technical architecture is differentiated. The market will likely support two D2D operators in the US (as it supports multiple MNOs), with network differentiation driving carrier selection. Outside the US, AST's multi-carrier global agreements represent a substantial lead over SpaceX's current international D2D footprint.

FCC Regulatory Environment

The FCC's 2024 Report and Order enabling supplemental coverage from space (SCS) provided the regulatory framework for D2D satellite services in licensed spectrum. This rule, while contested in some procedural dimensions, codified AST's operating model. A reversal would be a severe regulatory risk; absent that, the framework is established and AST is building within it.

Capital Markets Environment

ASTS has been a capital-intensive equity story since its 2021 SPAC merger, requiring multiple follow-on equity raises. The 2024–2025 period saw AST successfully raise capital at various price points, with the balance sheet backstopped by the DoD contracts and equity lines. However, the share count inflation from ~165M shares at the SPAC to 250M+ by mid-2025 has been a persistent dilution drag on per-share value accretion. The key forward signal is whether Block 2 commercial revenue can outpace the remaining CapEx funding requirement, reducing the probability of another large dilutive raise.

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4. Consumer & Public Opinion

AST SpaceMobile's B2B wholesale structure means end-consumer sentiment is largely irrelevant to the core investment thesis — subscribers never interact with "AST SpaceMobile" as a brand; they interact with AT&T, Verizon, or Vodafone. Negative or positive consumer sentiment about the MNO partners flows through to AST only insofar as it affects subscriber counts and MNO commercial health.

That said, the retail investor community around ASTS is exceptionally active and opinion-forming in ways that materially affect the stock's short-term trading dynamics. Reddit communities (particularly r/ASTS and r/SpaceMobile) and StockTwits maintain large, engaged retail investor bases that correlate satellite deployment updates with stock price moves. The announcement of Block 1 commercial go-live generated multi-day trading volume anomalies exceeding 5x average. This retail engagement creates both liquidity (tight spreads, responsive price discovery) and volatility risk (momentum-driven drawdowns on any execution stumble).

Public opinion on satellite connectivity broadly is favorable — rural broadband access is a bipartisan political priority in the US, and the FCC's Broadband Data Collection maps consistently show large rural coverage gaps that politicians across the aisle want addressed. This creates a supportive regulatory environment and positions AST as aligned with public-interest connectivity goals, which reduces headline regulatory risk.

The one consumer-facing dimension that matters to the thesis is MNO subscriber churn and satisfaction in coverage areas. Anecdotally, field tests and beta user reports for Block 1 service in the US have been positive on connectivity availability (the fact that service exists where terrestrial coverage does not) but mixed on throughput consistency under current single-digit satellite count. Block 2 will substantially improve this, but consumer-facing performance credibility for the carrier partners depends on execution.

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5. Key Risks & Structural Barriers

1. Execution Risk: Constellation Scale vs. Commercial Viability The most immediate and probability-weighted risk. Block 1 (5 satellites) provides limited coverage density and throughput. Block 2 (~60 satellites) is required for commercially meaningful service. Any launch failure, manufacturing delay, or satellite anomaly that substantially reduces Block 2 satellite count or delays deployment by 12+ months compresses the commercial revenue ramp directly and accelerates the need for additional capital. Space is unforgiving — a single major on-orbit failure of a large Block 2 satellite cannot be repaired. The probability of at least one partial mission failure in a 60+ satellite program is non-trivial; the question is whether AST has sized its constellation plan with sufficient redundancy.

2. Capital Structure and Dilution Risk AST's balance sheet has been funded primarily through equity rather than debt, given the pre-revenue profile. Each equity raise above current market prices is accretive; each raise below current prices is dilutive to existing holders. The company's cash runway, against a Block 2 manufacturing and launch budget of several hundred million dollars, creates ongoing binary risk around equity issuance timing and price. If the stock re-rates downward before the final Block 2 funding tranches, dilution could be severe. Public filings through Q2 2025 do not yet indicate full Block 2 funding closure — this remains an open variable.

3. Competitive Displacement by SpaceX/T-Mobile While SpaceX's D2D service does not immediately replicate AST's technical architecture or multi-carrier global agreements, SpaceX's execution track record, launch cost advantage (Starlink satellites ride their own rockets at near-zero marginal launch cost), and balance sheet are objectively superior to AST's. If SpaceX chooses to aggressively expand D2D carrier agreements globally and achieves competitive throughput with its next-generation satellites, AST's international first-mover advantage erodes faster than the current MOU pipeline would suggest. The risk is not displacement today — it is relative execution pace over the next 24–36 months.

4. MNO Partnership Concentration and Non-Binding Nature Several of AST's international carrier agreements are MOUs — memoranda of understanding — not binding commercial contracts with revenue commitments. An MNO that deprioritizes coverage-gap investment, changes strategy, or enters into a competing arrangement (e.g., a national carrier signing with SpaceX or with a regional D2D alternative) reduces AST's total addressable paying-subscriber base. The disclosed AT&T and Verizon agreements in the US are further along commercially, but terms of revenue sharing are not publicly disclosed in sufficient detail to model revenue per subscriber with confidence.

5. Spectrum Complexity in Emerging Markets Internationally, AST's value proposition depends on receiving regulatory approval to operate in each carrier's licensed spectrum in each jurisdiction. Regulatory timelines in emerging markets (sub-Saharan Africa, South/Southeast Asia, Latin America) are unpredictable and subject to local spectrum policy shifts. India, one of the largest potential markets, has been complex for satellite broadband entrants broadly. Any jurisdiction-by-jurisdiction spectrum delay reduces the subscriber addressable market in the near-to-medium term and pushes out revenue recognition.

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6. Investment Recommendation

Rating: Speculative Buy | Conviction: Moderate | Time Horizon: 18–30 months

ASTS represents a high-asymmetry, high-risk position in a winner-take-most infrastructure category. The technology works — Block 1 demonstrated that. The business model is sound — wholesale D2D connectivity to investment-grade MNOs is a superior revenue quality profile compared to direct consumer satellite services. The international agreement pipeline is real and differentiated from SpaceX's current footprint.

The recommendation is Speculative Buy rather than outright Buy for two reasons: (1) the stock is not optically cheap on any current-revenue multiple — it is priced on successful Block 2 deployment, and that deployment is subject to execution risk that cannot be diversified away; (2) the capital structure trajectory requires either commercial revenue acceleration or additional dilution, and both timing and price of any future equity raise are unknown. Position sizing should reflect that this is a real-option instrument, not a cash-flow compounder.

What would upgrade the call to Buy:

- Block 2 launches proceeding on schedule with confirmed on-orbit health by end of 2025
- At least one binding commercial revenue contract with disclosed per-subscriber economics
- Balance sheet evidence that Block 2 manufacturing and launch is fully funded without a large equity raise

What would downgrade the call to Sell/Avoid:

- A major on-orbit satellite failure reducing Block 2 operational capacity below commercial viability thresholds
- A large below-market equity raise ($10/share or below) that confirms capital desperation
- Binding SpaceX/T-Mobile international carrier agreements in AST's top-5 international markets

The long-term structural case for D2D satellite connectivity as a universal coverage layer is one of the more durable infrastructure theses in the current technology landscape. Whether AST specifically captures it depends on 24 months of execution that is binary in nature. Size accordingly.

Data sources: AST SpaceMobile public SEC filings (10-K/10-Q through Q2 2025), FCC license records, company press releases and investor presentations through August 2025, GSMA connectivity gap data, ITU broadband access statistics. All financial figures based on public disclosures; where figures are estimated or extrapolated from partial disclosures, this is indicated in context. This memo does not constitute investment advice.